

99.500 Applied Mathematics for Engineering

Homework 5

April 7, 2026

1. A semi-infinite slab is initially at a uniform temperature T_i . Starting at time $t = 0$, the surface temperature increases linearly with time, given by:

$$T(x = 0, t) = T_i + \alpha t \quad (1)$$

where α is a constant. Using the Laplace transform technique, derive an expression for the heat flux q at the surface of the slab as a function of time.

2. A solid cylinder is initially at a uniform temperature T_i . At time $t > 0$, the temperature at its outer surface where $r = r_o$ is suddenly changed to T_0 . Using the Laplace transform method, determine the temperature distribution $T(r, t)$ by solving the following heat conduction equation in cylindrical coordinates:

$$\frac{\partial T}{\partial t} = \beta \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) \quad (2)$$

where β is constant.